A single forking token fixes the algorithm a code model writes

Executive summary of the run's internal research report

26 October 2026

1. Headline

Where Qwen3-8B writes two algorithms for one task, a single generated token, the forking token, fixes which one it writes: on minimum spanning tree weight, trapped rain water and valid anagram the algorithm's entropy given that token's class is 0.050, 0.153 and **0.189 bits (95% interval 0.123 to 0.239)**, against 0.82 to 1.02 bits just before it (pre-registered bar 0.3 bits). Temperature acts only through that token's odds: on the same random streams, temperature 0.3 to 1.3 changed the algorithm in 0 of 20 seeds on each of three collapsed tasks, while a random push of the task-specification states changed it in more seeds, by **0.545 per seed (95% interval 0.41 to 0.67)**, pooled over shortest path and k-th smallest (40 seeds; one-sided sign-flip $p = 1.0 \times 10^{-5}$).

Source: the fork from `gen_art_experiment_35` (iteration 7); the contrast from the same seed experiment `gen_art_experiment_21` (iteration 4). **Evidence grade:** both experiments' verdicts are **PARTIAL**: the fork test passed on 3 of 3 units but the temperature prediction on 1 of 3; the contrast was pre-registered and held, but the plan-noise prediction (at least 5 of 20 seeds) failed on k-th smallest at 4.55. The round record grades both strands leads. **Scope:** Qwen3-8B, thinking off; forks on three mixed tasks, found blind on two held-out gemma-3-4b-it tasks but not on Llama-3.1-8B-Instruct's grid task; the contrast on two tasks in an 8-bit (FP8 W8A8) engine only (the push escapes alone reproduce in bfloat16). **Check:** the fork experiment's `results/test1.json` (0.0503, 0.1527, 0.1894 bits) and the same seed experiment's `results/c1seed.json` (mean 0.5449999999999999, interval 0.41375 to 0.6699999999999999, p $9.99990000099999 \times 10^{-6}$) hold these values and both PARTIAL verdicts; the paper states them the same way.

2. Goal

The run took as its starting point an inspiration repository in which a closed model wrote Dijkstra's algorithm in 125 of 125 programs for a weighted shortest path task and, under cumulative bans, moved to SPFA and Bellman-Ford rather than spreading over the alternatives. It asked how "creativity", meaning the choice of algorithm rather than its wording, shows up inside open-weight models and whether it can be influenced. The specific questions were these:

- Is sampling temperature the creativity knob under a shared seed, and do random context seeds or instructions change the algorithm?
- What does the internal geometry of algorithm choice look like (manifolds with geodesic distances and MDS, sparse autoencoders, circuits, natural language autoencoders, lenses)?
- Where does a ban of the top five algorithms send the model: to the edge of the banned set or to the opposite side?
- Can any knob produce a correct, truly novel algorithm?

The hypothesis entering the run was that a prohibition suppresses the default and promotes an associated substitute. Almost all work used Qwen3-8B; Llama-3.1-8B-Instruct, gemma-3-4b, Qwen2.5-7B-Instruct and larger Qwen3 and Gemma models were added as the run went on.

3. What was tried

From iteration 2 on, every iteration also opened with an audit that re-derived the previous iteration's numbers from the rows on disk.

1. **Prompt bans substitute, not just subtract** (7 artifacts). A screen of six accounts of what picks the algorithm after a ban (association, read-position manifold geometry, forking tokens, a decommitment direction, post-training, decoding knobs) on Qwen3-8B's grid and shortest path tasks, plus a source dataset for a coding bench.
2. **Bans remove the default, not pick the next** (7). The iteration-1 record was repaired. The iteration then built a likelihood instrument for a latent ladder, traced whether a ban acts through the prompt states or at generation, tried the top-five

prohibition, replicated in Llama, froze a 12-task bench and read the ban with SAEs.

3. **Plan nudges, not temperature, move ideas** (7). Ban-free forecasts of where bench bans land (Qwen3-8B and Llama), an exposure test (show a solution, ban it), the route of a ban per task, a search for a prohibition direction and locus, and how far bans and token vetoes reach.
4. **Plan nudges reopen what tuning narrowed** (7). Random "plan noise" pushes at task-specification states were priced against temperature and other knobs. The iteration also located where noise must enter and ran the same-seed contrast, then tested a plan subspace, a held-out family (gemma-3-4b-it) and its positioning against published work.
5. **A small hidden plan picks the algorithm** (7). Base-model plan states were transplanted into the instruct model. Two panels tested whether the base repertoire predicts which new tasks move. Plan noise was compared with text-level plan samplers and matched decoding, and the panels were repeated in Gemma.
6. **A hidden plan tilts one deciding token** (6). An FP8 against bfloat16 engine control; where a ban and an instruction are carried (task-state exchange); a one-token fork read with Gemma Scope 2 SAEs; and the plan read in words by a natural language autoencoder on Qwen2.5-7B-Instruct.
7. **One early token decides the algorithm** (7). A fitted plan subspace (DAS), the fork located in generation (the headline), temperature acting through the fork, the dose unit of plan noise, an evolutionary search with an evaluator, and a controlled DAS with a circuit in Gemma.
8. **Banned models rename their default algorithm** (7). Temperature applied only at the fork, one DAS protocol for two families, the circuit behind Qwen3-8B's fork, a fair test of Gemma's dial, a temperature law on held-out families, and the top-five prohibition across ten model configurations.
9. **Knobs reshuffle ideas, search finds new ones** (4). Compliance by name against compliance by mechanism under bans, the top-five landing on manifolds built with and without names, and where post-training sharpened the fork (base against instruct in three families).
10. **Knobs reweight ideas; search only recombines** (7). Evaluator-guided island searches with Qwen3-8B as the mutator on online bin packing (two implementations) and constructive TSP. Also prohibition read with an exact behavioural instrument, discipline against container on the program map, and which weights hold the default at the fork. The run then declared.

4. Key results

Grades are the strand states of the round record. A dagger marks a number that comes from the round record and that this summary did not find in the artifact's result files; every other number below was found there.

Result	Numbers	Source (round)	Grade
Push against temperature on the same seeds	0.545 [0.41, 0.67], p 1.0×10^{-5}; push 17.25 and 4.55 of 20 seeds; temperature 0 of 20 per task	`gen_art_experiment_21` (4)	lead; PARTIAL
Temperature is a phrasing knob	797 of 800 programs BFS†; phrasing clusters 4 to 36†	`gen_art_experiment_6` (1)	lead
The push escapes are a model property, not 8-bit rounding	escapes, bfloat16 against FP8: connected components 0.65 vs 0.725, shortest path 0.15 vs 0.20	`gen_art_experiment_29` (6)	lead
A temperature law over fork logits	passes 5 of 6 units†, including held-out Gemma (0.033, 0.014); Llama fails (0.142); beats the flat law on 4 of 6† (sign test p 0.34)	`gen_art_experiment_43` (8)	lead
The forking token (headline), and its margin on collapsed tasks	one-token fork on 3 of 3 mixed units, entropy given the token at most 0.189 bits; collapsed fork margins from 4.85† up to 17.04 nats (connected components)	`gen_art_experiment_35` (7)	lead
Rename gap under a five-name ban (grid)	lexical minus mechanistic compliance 0.70 (runtime label) and 0.61 (tracer), pooled; positive in 10 of 10 configurations†; the tracer missed its gate	`gen_art_experiment_45` (9)	lead
Rename gap, exact instrument (bin packing)	Qwen3-8B at $k = 5$: +0.40 [0.21, 0.55], +0.08 on held-out templates; thinking models obey (+0.04, -0.13)†; 0 of 1,143 programs novel and better than best fit†	`gen_art_experiment_54` (10)	lead
Evaluator-guided search on bin packing	held-out excess at T 1.6: 1.40% / 3.55% against best fit's 3.98% / 5.37%; never without selection	`gen_art_experiment_51` (10)	lead
Replication search	0 of 40 judged programs novel†; DTB/RPB 0.74	`gen_art_experiment_52` (10)	lead
Where post-training wrote the default	base fork rows close 0.548 of the connected-components gap against 0.044 for isotropic edits; residual-subspace random edits match (p 0.37)	`gen_art_experiment_56` (10)	lead

5. What failed

Dead ends, each with what killed it (artifact and round):

- **Forecasting the substitute.** Ban-free forecasts on the bench failed: no forecast family beat uniform (ladder +0.06, association −1.78), and 3 of 5 landings were unforeseen (`gen_art_experiment_13`, round 3; null). The iteration-1 association map had named 1 of 2 substitutes on two events, so the ladder was never tested there (`gen_art_experiment_1`, round 1).
- **Manifold proximity as the rule.** The ban shift lay 61–99% off the mode axes, and affine ablation moved 0 of 50 (`gen_art_experiment_2`, round 1; null). The fitted plan subspace was disconfirmed: it ranked fresh directions at ρ 0.03, and removing it kept the effect (`gen_art_experiment_22`, round 4; null).
- **A single decommitment or prohibition direction.** The decommitment direction moved Hill by +0.65, the same as a random direction's +0.59 (`gen_art_experiment_4`, round 1; null). A shared prohibition direction did no better than random ones (0 of 5); that strand is graded a lead only for the random pushes it found instead (`gen_art_experiment_17`, round 3).
- **Likelihood instruments.** The FRLP gates failed: the effective sample size was below 5 in 98% of cells in one experiment and had a median of 2.5 in the other (`gen_art_experiment_7` and `gen_art_experiment_9`, round 2; broken). The SAE transfer failed (dCE ratio 1.59–3.29; `gen_art_experiment_12`, round 2; null).
- **The base model's repertoire as a predictor, and transplants.** The transplant was disconfirmed in 1 of 5 (`gen_art_experiment_24`, round 5). Movability ρ was −0.03 and 0.21 (`gen_art_experiment_25` and `gen_art_experiment_26`, round 5).
- **A certified variable for the algorithm.** Matched DAS was disconfirmed: label controls and untrained networks kept much of the effect, and Qwen3-8B's fork has no handle (`gen_art_experiment_40`, round 8; null). Token walks stalled in 6 of 6 (`gen_art_experiment_18`, round 3; null).
- **Fork-targeted search operators.** On TSP the operator's kill condition was met (DTB/T1.0 0.96, DTB/RPB 0.95), and no best program beat the ReEvo/EoH seed rule at 13.67% (`gen_art_experiment_53`, round 10; null).
- **Novelty.** No knob, prompt, internal edit or prohibition produced a correct novel algorithm. The eight bin-packing programs one reader classed as new were not reproduced by the independent replication.
- **The source dataset** delivered data but no frozen bench (`gen_art_dataset_1`, round 1; broken).

6. Open questions

- Whether the whole-multiple-of-item bin-packing rule is new. Settling it needs an ablation of the term, a second blind reader, and a comparison with the published target-residual programs.
- Where a search should spend its randomness once the prompt is fixed. One instruction sentence decided whether temperature explored or copied.
- Why the same reasoning model renames a banned algorithm on the grid but obeys on bin packing.
- Whether a variable for the algorithm exists anywhere in the model. Specificity failed at the late fork sites under every method tried, and at the readout rows against random edits in the residual subspace.
- Whether the push-against-temperature contrast holds in bfloat16 and on more tasks and models; it was measured in one 8-bit engine on two tasks.
- A program map that separates algorithmic discipline from container and surface, built from programs by several authors.

The full chronological report is `report.pdf`, 471 pages, with every table typeset in the iteration that produced it.